(a) Harmonic Nodal Chambers $\Omega_1, \ldots, \Omega_{25}$ & Euler–Stirling Boundary $d_{\mathrm{Euler}}(t)$

Spectral height t

Transverse offset $d = \sigma - 1/2$

Scaled imaginary field $e^{\pi t/4} I(d, t)$ $(\Delta I = 0)$

Imaginary nodal walls Γ_k $(I(d, t) = 0$ from $\tau_k)$

Real zero curves $\mathcal{R}_k$ $(R(d, t) = 0$ from $\gamma_k)$

Euler–Stirling right boundary $d = d_{\mathrm{Euler}}(t)$

Boundary of critical strip $d = 1/2$ $(\sigma = 1)$

Riemann zeros $\gamma_1, \ldots, \gamma_{25}$ on $d = 0$

Critical-line extrema $\tau_1, \ldots, \tau_{25}$ $(\Xi'(\tau_k) = 0)$

(b) Chambers $\Omega_1, \ldots, \Omega_5$ & Parabolic Curvature $t(d) - \gamma_k = \frac{1}{2}\kappa(\gamma_k) d^2 + \mathcal{O}(d^4)$

Ω_5: $I > 0$

Ω_4: $I < 0$

Ω_3: $I > 0$

Ω_2: $I < 0$

Ω_1: $I > 0$

Nodal walls Γ_k $(I = 0,\ t'(0) = 0)$

Exact zero curves $\mathcal{R}_k$ $(R = 0)$

Parabolic law $\gamma_k + \frac{1}{2}\kappa(\gamma_k) d^2$

Spectral height t

Transverse offset $d = \sigma - 1/2$

(c) Cauchy–Riemann Monotonicity $\frac{d|R|}{ds} = |\nabla I| = |\Xi'(z)| > 0$ Along $\Gamma_1, \ldots, \Gamma_{25}$

Relative amplitude $|R(d, t_{\Gamma_k}(d))|/|\Xi(\tau_k)|$ (log scale)

Γ_1 $(\tau_1 = 15.6)$

Γ_5 $(\tau_5 = 34.2)$

Γ_{10} $(\tau_{10} = 50.8)$

Γ_{15} $(\tau_{15} = 65.8)$

Γ_{20} $(\tau_{20} = 78.0)$

Γ_{25} $(\tau_{25} = 90.0)$

Arrival at $d = d_{\mathrm{Euler}}(t_k)$

Critical-line base $|R(0, \tau_k)|/|\Xi(\tau_k)| = 1$

Transverse offset $d = \sigma - 1/2$

(d) Parabolic Zero-Curve Curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$ Across $\gamma_1, \ldots, \gamma_{15}$

Vertex curvature $\kappa(\gamma_k) = t''(0) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$

Parabolic Taylor coefficient $c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$

γ_1: -1.16

γ_4: -1.91

γ_5: -0.93

γ_8: -0.90

γ_9: -2.40

γ_{10}: -0.85

γ_{13}: -2.25

γ_{14}: -0.29

γ_{15}: -2.28

Curvature $\kappa(\gamma_k)$ and coefficient $c_{\mathrm{par}}(\gamma_k)$

Non-trivial Riemann zero index k ($\gamma_1 \approx 14.13$ to $\gamma_{15} \approx 65.11$)